

Q1:- Use the ftn to prove that (a) $f_x(0,0)$ & $f_y(0,0)$ exist (by def) & (b) f is not differentiable at $(0,0)$.

$$* f(x,y) = \begin{cases} \frac{3x^2y}{x^4+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

Sol

$$(a) f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x}$$
$$= \lim_{\Delta x \rightarrow 0} \frac{0 - 0}{\Delta x} = 0$$

$$f_y(0,0) = \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0,0)}{\Delta y}$$
$$= \lim_{\Delta y \rightarrow 0} \frac{0 - 0}{\Delta y} = 0$$

So the partial derivatives at $(0,0)$ exists.

(b) f is not differentiable at $(0,0)$ by showing that it is not continuous at this pt. To see that f is not continuous at $(0,0)$,
Along the line $y = mx$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{Along } y=mx}} f(x,y) = \frac{3x^2 \cdot mx}{x^4 + m^2x^2}$$

$$\lim_{x \rightarrow 0} f(x) = \frac{3mx^3}{x^2[x^2+m^2]}$$

$$\lim_{x \rightarrow 0} f(x) = \frac{3mx}{x^2+m^2}$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{3mx}{x^2+m^2} = 0$$

Along $y = mx^2$
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \frac{3x^2 \cdot mx^2}{x^4 + m^2 x^4}$
 Along $y = mx^2$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{3mx^4}{x^4[1+m^2]}$$

$$= \frac{3m}{1+m^2} \quad \text{Path dep}$$

So the limit of $f(x,y)$ as $(x,y) \rightarrow (0,0)$ does not exist.

So we can say that f is not continuous at $(0,0)$. So according to theorem we can say that f is not diff at $(0,0)$.

$$* f(x,y) = \begin{cases} \frac{2x^2 y^2}{x^4 + y^4} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

Sol (a) $f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x}$

$$= \lim_{\Delta x \rightarrow 0} \frac{0-0}{\Delta x}$$

$$= 0$$

$$f_y(0,0) = \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0,0)}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{0-0}{\Delta y}$$

$$= 0$$

So partial derivatives at $(0,0)$ exists.

(b) f is not diff at $(0,0)$ by showing that it is not

Continuous at this pt. To see that f is not cont at $(0,0)$, we find limit

Along line $y = mx$

$$\lim_{(x, mx) \rightarrow (0,0)} f(x, mx) = \lim_{x \rightarrow 0} \frac{2x^2 \cdot m^2 x^2}{x^4 + m^4 x^4}$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{2m^2 x^4}{x^4 [1 + m^2]}$$

$$= \frac{2m^2}{1 + m^2} \quad \text{Path dep}$$

So the limit of $f(x, y)$ as $(x, y) \rightarrow (0, 0)$ does not exist.
So we can say that f is not continuous at $(0, 0)$.
So according to theorem we can say that f is not diff at $(0, 0)$.

Q2:- Consider $f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$

Show that $f_x(0, 0)$ & $f_y(0, 0)$ exist (by def) but f is not continuous at $(0, 0)$.

Sol $f_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(0 + \Delta x, 0) - f(0, 0)}{\Delta x}$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{0 - 0}{\Delta x}$$

$$= 0$$

$$f_y(0,0) = \lim_{\Delta y \rightarrow 0} \frac{f(0,0+\Delta y) - f(0,0)}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{0-0}{\Delta y}$$

$$= 0$$

For Continuity $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = f(0,0)$

Along $y = mx$

$$\lim_{(x,mx) \rightarrow (0,0)} f(x,mx) = \frac{x \cdot mx}{x^2 + m^2 x^2}$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{mx^2}{x^2[1+m^2]}$$

$$= \frac{m}{1+m^2} \text{ Path dep}$$

As limit does not exist so f is not Continuous at $(0,0)$.

Q3:- Show that f has partial derivatives at $(0,0)$. Also show that f is not Continuous at $(0,0)$. Is f diff at $(0,0)$.

$$f(x,y) = \begin{cases} \frac{xy(1+y^2)}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

Sol $f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(0+\Delta x, 0) - f(0,0)}{\Delta x}$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta x \cdot 0 (1+0) - 0}{\Delta x} = 0$$

$$\begin{aligned}
 f_y(0,0) &= \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0,0)}{\Delta y} \\
 &= \lim_{\Delta y \rightarrow 0} \frac{0 \cdot \Delta y (1 + \Delta y^2) - 0}{\Delta y} \\
 &= 0
 \end{aligned}$$

f has P.D's at $(0,0)$.

For Continuity at $(0,0)$ $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = f(0,0)$

Along $y = mx$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } y=mx}} f(x,y) = \frac{x \cdot mx (1 + m^2 x^2)}{x^2 + m^2 x^2}$$

$$\begin{aligned}
 \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{mx^2 (1 + m^2 x^2)}{x^2 (1 + m^2)} \\
 &= \lim_{x \rightarrow 0} \frac{m (1 + m^2 x^2)}{1 + m^2} \\
 &= \frac{m}{1 + m^2} \quad \text{Path dep}
 \end{aligned}$$

So f is not Continuous at $(0,0)$ as limit does not exist. f is not Continuous at $(0,0)$ so it is not diff. at $(0,0)$.

$$\lim_{x \rightarrow 0} \frac{3m^2 x^4}{x^4 + m^4 x^4}$$

$$\lim_{x \rightarrow 0} \frac{x^4 [3m^2]}{x^4 [1+m^4]}$$

$$\frac{3m^2}{1+m^4} \text{ Path dep}$$

Limit does not exist

Q10:- (a) Determine whether each ftn f is Continuous at $(0,0)$

(b) If f is discontinuous at $(0,0)$, is it possible to define $f(0,0)$ so that f would be Continuous at $(0,0)$?

(c) If the answer to (b) is yes, how should $f(0,0)$ be defined?

$$(i) f(x,y) = \frac{xy^2}{x^2+y^2}$$

Sol Along $y = mx$

$$\lim_{(x,mx) \rightarrow (0,0)} \frac{x \cdot m^2 x^2}{x^2 + m^2 x^2}$$

$$\lim_{x \rightarrow 0} \frac{m^2 x^3}{x^2 + m^2 x^2}$$

$$\lim_{x \rightarrow 0} \frac{m^2 x^3}{x^2 [1+m^2]}$$

$$\lim_{x \rightarrow 0} \frac{m^2 \cdot x}{1+m^2}$$

0

Along $y = mx^2$

$$\lim_{(x,mx^2) \rightarrow (0,0)} \frac{x \cdot m^2 x^4}{x^2 + m^2 x^4}$$

$$\lim_{x \rightarrow 0} \frac{m^2 x^5}{x^2 [1+m^2 x^2]} \Rightarrow \lim_{x \rightarrow 0} \frac{m^2 x^3}{1+m^2 x^2} = 0$$

The ftn is discontinuous at $(0,0)$. Yes it is possible to define $f(x,y)$ at $(0,0)$. $f(0,0)$ should be 0.

$$(ii) f(x,y) = \begin{cases} \frac{3xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

sol Along $y = mx$

$$\lim_{(x,mx) \rightarrow (0,0)} \frac{3x \cdot mx}{x^2 + m^2 x^2}$$

$$\lim_{x \rightarrow 0} \frac{3mx^2}{x^2[1+m^2]}$$

$$\frac{3m}{1+m^2} \text{ (Path dep)}$$

$$f(0,0) = 0$$

As the limit does not exist so ftn is discontinuous at $(0,0)$. No, it is not possible to define $f(x,y)$ at $(0,0)$.

Q11:- Find each limit by converting to polar coordinates

$$(a) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$$

sol Let $x = r \cos \theta$, $y = r \sin \theta$, $x^2 + y^2 = r^2$

$$\lim_{r \rightarrow 0} \frac{r^2 \cos^2 \theta \cdot r \sin \theta}{r^2 \cos^2 \theta + r^2 \sin^2 \theta}$$

$$\lim_{r \rightarrow 0} \frac{r^3 \cos^2 \theta \sin \theta}{r^2 [\cos^2 \theta + \sin^2 \theta]}$$

$$\lim_{r \rightarrow 0} r \cos^2 \theta \sin \theta$$

$$= 0$$

Q16 :- Show that the fcn

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

has partial derivatives at $(0, 0)$, but is ~~not~~^{not} continuous at $(0, 0)$.

$$\underline{\text{sol}} \quad f_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(0 + \Delta x, 0) - f(0, 0)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\frac{0}{\Delta x^2} - 0}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{0}{\Delta x}$$

$$= 0$$

$$f_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{f(0, 0 + \Delta y) - f(0, 0)}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0, 0)}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{\frac{0}{\Delta y^2} - 0}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{0}{\Delta y}$$

$$= 0$$

The partial derivatives $f_x(0,0)$ & $f_y(0,0)$ both exist

Along $y = mx$

$$\lim_{(x, mx) \rightarrow (0,0)} \frac{x \cdot mx}{x^2 + m^2 x^2}$$

$$\lim_{x \rightarrow 0} \frac{mx^2}{x^2[1+m^2]}$$

$$\frac{m}{1+m^2} \quad (\text{Path dep})$$

Limit does not exist. So f is not continuous at $(0,0)$.